

Homework 6: Fourier Methods and Transformations

This homework covers material from **Module 6: Fourier Methods, Transformations, and Data Enhancement** and aligns with learning outcomes **LO6, LO8**.

Related Resources: - Lectures: M6 lecture notes (Fourier-Domain Modeling, Data Transformations) - Lab: Lab 6 (Fourier Filtering and Transformations) — *Essential for computational Problems 6.2, 6.3, 6.4* - Lab 6.5: Continuation Techniques (related to Problem 6.2) - Textbook: Blakely Ch. 11, 12 - Python libraries: NumPy FFT, SciPy signal processing

Problems

1. **Problem 6.1** (Blakely Ch. 11, Problem 1) — *Fourier Analysis/Moderate*

- Apply the Fourier transform to the potential of a vertical line source. Show how the wavenumber relates to source geometry.
- **Cross-reference:** Lab 6, Part 1; M6 lecture notes, Section “Fourier-Domain Modeling”; M0/lecture2_notes.qmd (FFT review)
- **Difficulty:** Moderate (Fourier transform mechanics; wavenumber interpretation)
- **Tip:** Use linearity of FT; integrate in k-domain; relate slope to source depth, amplitude to mass/moment

2. **Problem 6.2** (Blakely Ch. 12, Problem 2) — *Continuation/Challenging*

- Derive the upward continuation operator in the wavenumber domain $e^{-|k|h}$. Discuss stability for downward continuation.
- **Cross-reference:** Lab 6, Part 2; Lab 6.5; M6 lecture notes, Section “Transformations”; M6.5 lecture notes
- **Difficulty:** Challenging (separation of variables in Laplace’s eq.; operator derivation)
- **Tip:** Start with Laplace’s equation in Fourier domain; solve for spectral expansion; discuss exponential amplification in downward case

3. **Problem 6.3** (Blakely Ch. 12, Problem 4) — *Transformation/Challenging*

- Implement reduction to the pole for a magnetic anomaly with known magnetization direction. Show the filter in wavenumber space.
- **Cross-reference:** Lab 6, Part 3; M6 lecture notes, Section “Phase Transformations”; M3/lecture notes (magnetization)
- **Difficulty:** Challenging (phase rotation; filter design; physical interpretation)
- **Tip:** Apply analytic signal to estimate inclination/declination; implement RTP filter; verify with synthetic data before applying to real data

4. **Problem 6.4** (Blakely Ch. 12, Problem 6) — *Signal Processing/Moderate*

- Calculate the analytic signal of a magnetic profile using the Hilbert transform. Explain how it enhances source edges.

- **Cross-reference:** Lab 6, Part 4; M6 lecture notes, Section “Complex Analysis and Analytic Signal”; M0/lecture2_notes.qmd (signal processing)
- **Difficulty:** Moderate (Hilbert transform computation; edge detection concept)
- **Tip:** Use `scipy.signal.hilbert` or manual FFT implementation; plot amplitude and phase; relate to source geometry

5. **Problem 6.5** (Blakely Ch. 11, Problem 5) — *Spectral Analysis/Challenging*

- Interpret the power spectrum of a gravity anomaly. How does the slope relate to source depth and geometry?
- **Cross-reference:** Lab 6, Part 1; M6 lecture notes, Section “Depth and Shape from Spectra”; M4/homework problem 4.5 (harmonic analysis analogy)
- **Difficulty:** Challenging (spectral interpretation; inverse modeling)
- **Tip:** Compute FFT magnitude; plot log-log spectrum; identify break frequency; relate slope to fractal dimension; estimate equivalent source depth

Submission Guidelines

- Submit solutions as a PDF, Quarto document, or **Jupyter notebook** (preferred for computational work)
- Include complete derivations and mathematical justifications
- **Essential:** Include plots of filters (wavenumber domain), spectra (log-log plots), and before/after transformed data
- Include Python code or pseudocode for computational problems
- Discuss physical meaning of results and limitations
- Expected complexity: 8–12 pages + extensive figures/code